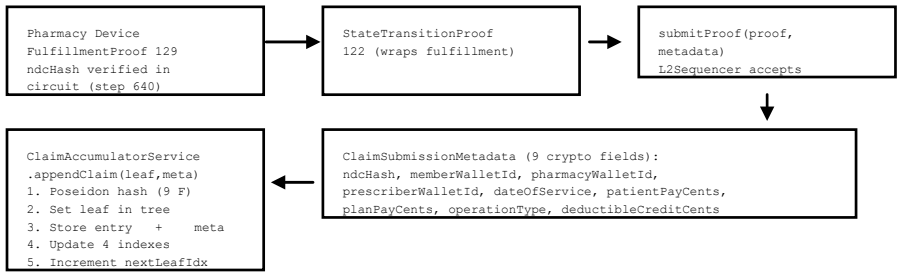
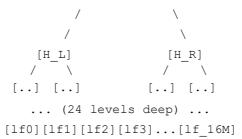
FIG. 9 -- CLAIM ACCUMULATOR TREE WITH FOUR PARALLEL INDEXES 900

SECTION A: DATA FLOW -- ZK PROOF -> CLAIM LEAF 900



SECTION B: MERKLE TREE(24) STRUCTURE 902



ClaimLeaf (9-Field Poseidon Hash):

leafHash = Poseidon.hash([
F1: memberWalletId, F2: ndcHash,
F3: pharmacyWalletId, F4: prescriberWalletId,
F5: dateOfService, F6: patientPayCents,
F7: planPayCents, F8: operationType,
F9: deductibleCreditCents
])
Capacity: 2^24 = 16,777,216 leaves
All identity fields = Poseidon hashes. No plaintext PHI.

SECTION C: FOUR PARALLEL INDEX STRUCTURES 904

Each index: Map<string, bigint[]> Key = Poseidon hash Value = leaf indices

INDEX 1: memberIndex
Map<memberWalletId, bigint[]>

```
member_A -> [0, 5, 12]
member_B -> [1, 8]
member_C -> [3, 7, 9]
```

INDEX 2: pharmacyIndex
Map<pharmacyWalletId, bigint[]>

```
pharm_X -> [0, 1, 3, 5]
pharm_Y -> [7, 8, 9, 12]
pharm_Z -> [2, 4, 6]
```

INDEX 3: prescriberIndex
Map<prescriberWalletId, bigint[]>

```
prescriber_P -> [0, 1, 3]
prescriber_Q -> [5, 7, 8]
prescriber_R -> [2, 4, 6]
```

INDEX 4: ndcIndex
Map<ndcHash, bigint[]>

```
ndc_Atorva -> [0, 5]
ndc_Metform -> [1, 8, 12]
ndc_Amoxi -> [3, 7, 9]
```

Single claim at leafIndex N appears in ALL FOUR indexes (member, pharmacy, prescriber, ndc)

SECTION D: QUERY AND MERKLE WITNESS FLOW 906

6 RPC Methods: Merkle Witness Verification:
l2_getClaimsByMember(walletId) l2_getClaimWithProof(leafIndex):
l2_getClaimsByPharmacy(walletId) 1. entry = entries.get(leafIndex)
l2_getClaimsByPrescriber(walletId) 2. witness = tree.getWitness(idx) -- 24 siblings
l2_getClaimsByNdc(ndcHash) 3. root = tree.getRoot()
l2_getClaimWithProof(leafIndex) Verifier: computedRoot = witness.calculateRoot(
-> returns entry + 24-sibling witness Poseidon.hash([9 fields]))
l2_getClaimAccumulatorRoot() assert computedRoot == root -> PROVES claim exists
-> returns { root, leafCount } ClaimAccumulatorWitness extends MerkleWitness(24)

SECTION E: BATCH INTEGRATION 908

```
L2Receipt { claimLeafIndex?: bigint }
Links receipt to specific claim in tree
```

```
L2Batch { claimAccumulatorRoot?: string }
Snapshot of tree root at batch creation
```